TECHNOLOGICAL UNIVERSITY (THANLYIN)

DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

DETECTING FOR FACIAL REGION MODIFICATIONS USING DEEP LEARNING APPROACH

BY

MA MYA THINZAR KYAW

VI IT - 3

GRADUATION THESIS

OCTOBER, 2025

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We certify that we have examined, and recommend to the University Steering Committee for Bachelor of Engineering Graduation thesis entitled: **"DETECTING FOR FACIAL REGION MODIFICATIONS USING DEEP LEARNING APPROACH"** submitted by **MA MYA THINZAR KYAW**, Roll No. **VI-IT 3 (NOVEMBER, 2024)** to the Department of Information Technology Engineering as the requirement for the Bachelor of Engineering degree, **BE(IT)**.

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ABSTRACT

Digital image manipulation has become increasingly widespread with the advancement of artificial intelligence and photo-editing tools. Techniques such as face swapping and facial feature modification are commonly used across entertainment, social media, and in some cases, for malicious purposes. These manipulations are often subtle and difficult to detect visually, making the development of reliable detection systems an important area of research for security, digital forensics, and media verification. Existing deep learning-based detection approaches usually demand large datasets and high computational resources, which limits their practicality in lightweight applications.

This study presents a lightweight facial region modification detection system that leverages landmark-based analysis combined with similarity metrics. MediaPipe Face Mesh is used to detect 468 facial landmarks, enabling the precise segmentation of key facial regions such as the eyes, nose, and mouth. The Structural Similarity Index (SSIM) is then applied to compare corresponding regions between original and modified images. By highlighting the detected modifications, the system provides an interpretable output for users, ensuring both efficiency and accessibility without the need for large-scale training.

The proposed system was implemented using Python with libraries including OpenCV, MediaPipe, scikit-image, and Tkinter for GUI development. Experimental results demonstrate that the approach effectively identifies subtle modifications in facial regions and provides clear visual feedback through an intuitive interface. Although performance may decrease with low-quality or non-frontal images, the system proves to be well-suited for small-scale forensic applications and rapid media authenticity verification.

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